



BRAC UNIVERSITY
Department of Computer Science & Engineering
FINAL EXAMINATION, SPRING 2024
CSE 350: Digital Electronics and Pulse Techniques



Total Marks: 60 (20×3)

Time: **90 Minutes**

- Total no of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet
- There is no need to draw circuits in your answer sheet.

Name:	ID:	Section:
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Question 1

For this **Triangular Wave Generator** circuit, Given: $V_H = 12\text{ V}$, $V_L = -8\text{ V}$, $R_1 = 6\text{ k}$, $R_2 = 25\text{ k}$, $C = 1\text{ }\mu\text{F}$, $f = 5\text{ kHz}$ [20]

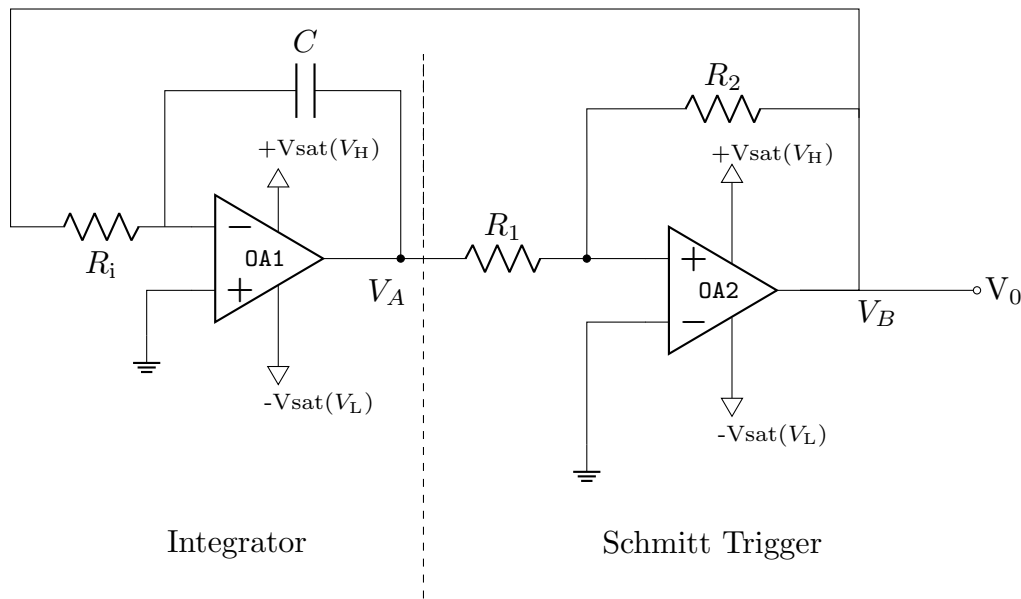


Figure 1

CO3	(a)	Determine the Duty Cycle.	[4]
CO3	(b)	Draw the waveforms of V_A and V_B vs time in the same plot with appropriate labels (must include values of V_H , V_L , V_{TH} , V_{TL} , T_1 , T_2 , T).	[6]
CO3	(c)	Calculate the value of R_i .	[10]

Question 2

The figure below indicates the operation of a non-uniform Flash ADC (Analog-to-Digital Converter). A standard binary-weighted DAC (Digital-to-Analog Converter) has been connected to the outputs of the ADC. [20]

CO3	(a)	Determine what should be the reference voltage of the ADC.	[2]
CO3	(b)	Design, draw and label all the part values of the Flash ADC which follows the transfer curve shown below.	[12]
CO3	(c)	Draw the DAC mentioned in the description with the same reference voltage found in (a). Find the analog outputs from the DAC.	[2+4]

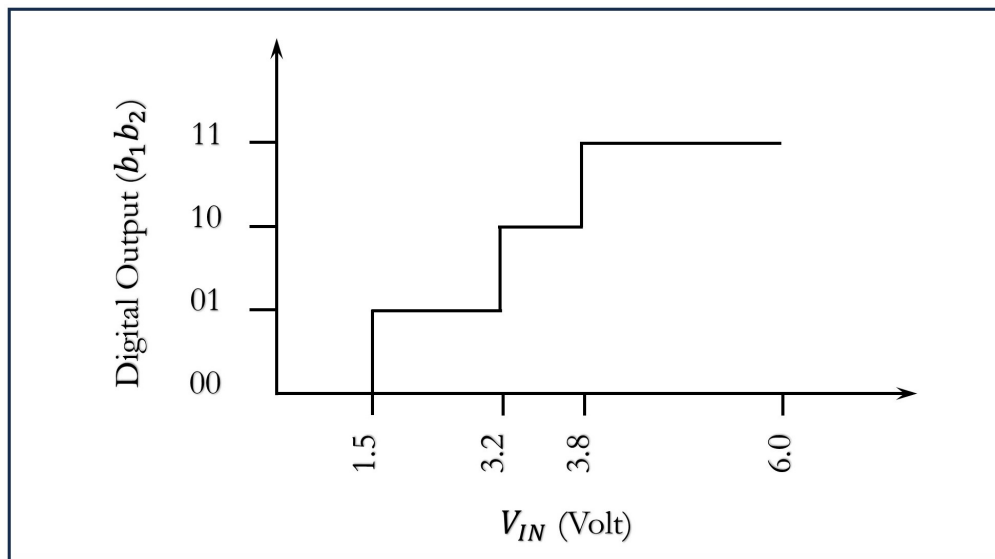


Fig : Transfer curve of a Flash ADC

Figure 2

Question 3

For this circuit, the hysteresis width is 5V and the value of R_x is $2.4\text{k}\Omega$. Consider the following two cases:

Case 1: When the switch is in position A and the value of the input voltage V_I is a very low negative value, the voltage at V_X is 5.2V.

Case 2: When the switch is in position B and the value of the input voltage V_I is a very high positive value, the voltage at V_X is 1.4V. [20]

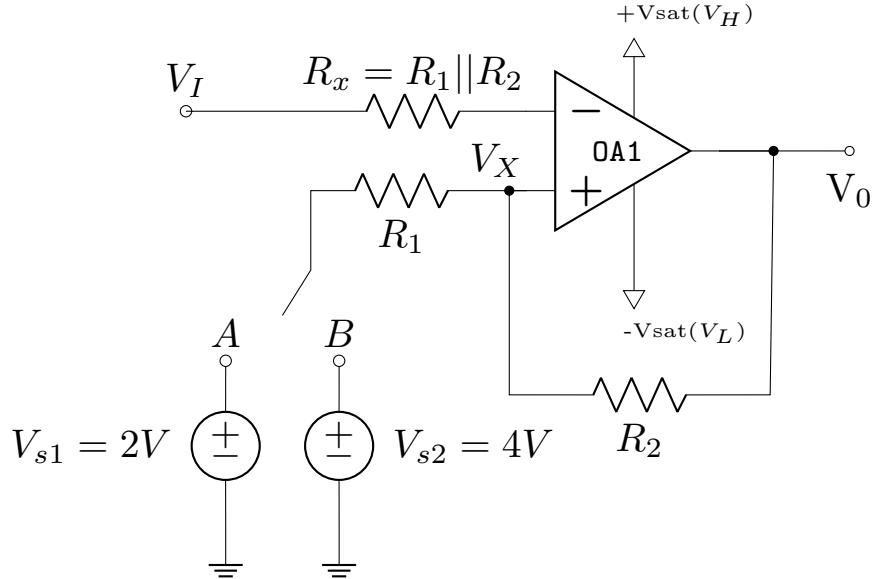


Figure 3

CO1	(a)	Find the value of R_1 and R_2 using only the information given above.	[12]
CO1	(b)	How many different voltages can appear at V_o ? Find the values of those voltages.	[4]
CO1	(c)	Draw the input vs output characteristics (V_I vs V_o) when the switch is connected to position A. [Clearly indicate all the values in the graph]	[4]

Question 4

The NMOS NOR gate in **figure 4** is biased at $V_{DD} = 5V$. The transistor parameters are $K_n = 0.3 \text{ mA/V}^2$, $V_{TN1} = 0.5V$, $V_{TN2} = 0.8V$.
For triode mode, $i_D = k_n [2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2]$, and for saturation mode, $i_D = k_n [V_{GS} - V_{TN}]^2$ [20]

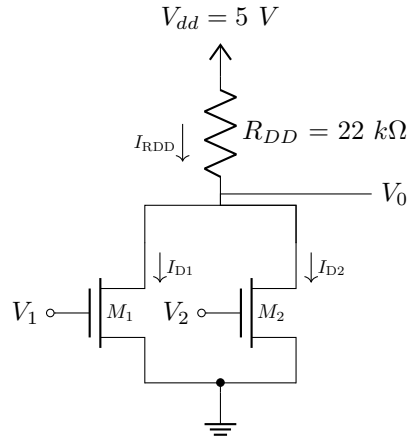


Figure 4: Circuit

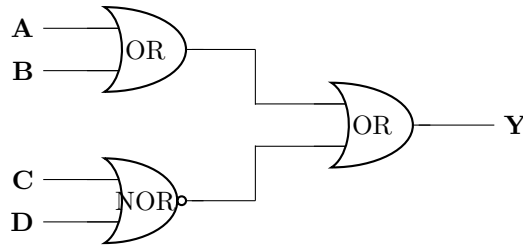


Figure 5

CO1	(a)	Find the voltage, V_0 , for all input combinations and make a truth table for the circuit in figure 4 .	[6]
CO1	(b)	Identify the boolean function, Y , of the logic circuit in figure 5 .	[2]
CO1	(c)	Hence, design the CMOS circuit for Y . (Clearly show the PMOS and NMOS networks).	[8]
CO1	(d)	Now, design circuit implementing the same boolean function, Y , from part (b) using multiple copies of the circuit in figure 4 only.	[4]